

Why the Octic Frontier Matters

Recoverability, Complexity, and the Complex Reserve-Currency Manifold

Soumadeep Ghosh with ChatGPT

Kolkata, India

Abstract

This paper investigates the relationship among reserve-currency architectures, monetary recoverability, latent-factor monetary systems, and the geometry of complex reserve-currency manifolds. We distinguish between mathematical admissibility and economic admissibility and argue that recoverability constraints impose powerful restrictions on the set of feasible monetary trajectories.

Starting from a bounded reserve-currency architecture satisfying

$$R_{\max} \leq 5,$$

we develop a quintic representation of reserve-currency demand and supply, derive a unique real equilibrium under perfect-demand and perfect-supply conditions, and examine the consequences of extending the reserve-currency manifold into the complex domain.

While the complex extension is mathematically natural, we show that recoverability constraints, behavioral asymmetries arising from loss aversion, geopolitical realities, military-security requirements, and institutional limitations dramatically reduce the economically admissible state space. We interpret the octic recoverable-money polynomial as a solvability frontier and demonstrate why movement toward a decic recoverable-money architecture introduces additional complexity, additional reserve-currency interactions, and potentially additional civilizations or monetary blocs.

The resulting framework connects monetary economics, finance, psychology, philosophy, political economy, international relations, warfare economics, statistics, artificial intelligence, representation learning, and systems theory.

The paper ends with “The End”

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1 Introduction

Modern monetary theory traditionally studies:

1. Currency production,
2. Currency circulation,
3. Currency management,
4. Currency withdrawal.

However, reserve currencies compete with:

- Goods,
- Commodities,
- Strategic inventories,
- Equities,
- Sovereign debt,
- Land,
- Foreign currencies,
- Strategic military reserves.

A modern theory of reserve currencies must therefore explain not merely how money is created but why economic agents voluntarily hold it.

The present paper advances three central propositions:

1. Reserve-currency demand emerges from latent structural factors.
2. Recoverability constraints govern admissible monetary trajectories.
3. The complexity of the reserve-currency manifold increases dramatically beyond the octic frontier.

2 The Five-Currency Reserve Architecture

Assume

$$R_{\max} \leq 5.$$

Definition 1. *The reserve-currency state vector is*

$$c = (c_1, c_2, c_3, c_4, c_5).$$

Each coordinate represents a reserve-currency dimension.

The reserve-currency manifold is

$$\mathcal{M}_5 = \{c : c \in \mathbb{R}^5\}.$$

Remark 1. *The five-currency hypothesis does not require five active reserve currencies.*

It merely asserts that no more than five reserve-currency dimensions are economically sustainable.

3 Reserve-Currency Demand and Supply

Define

D_i = Demand for reserve currency i ,

and

S_i = Supply of reserve currency i .

Demand equilibrium satisfies

$$F_D(x) = \sum_{i=1}^5 ((D_i - x)^5 - D_i^5) = 0.$$

Supply equilibrium satisfies

$$F_S(y) = \sum_{i=1}^5 ((S_i - y)^5 - S_i^5) = 0.$$

Theorem 1. *Each equilibrium equation possesses at most five real roots.*

Proof. Each equilibrium equation is a quintic polynomial.

A quintic polynomial possesses at most five real roots.

□

4 Perfect Demand and Perfect Supply

Consider

$$F_D(x) = 0,$$

$$F_S(y) = 0,$$

and

$$x^2 + y^2 = 0.$$

Theorem 2. *The system possesses exactly one real solution.*

Proof. Because

$$x^2 + y^2 = 0,$$

and both squares are nonnegative,

$$x = 0, \quad y = 0.$$

Substituting into the demand equation yields

$$F_D(0) = 0.$$

Substituting into the supply equation yields

$$F_S(0) = 0.$$

Hence

$$(x, y) = (0, 0)$$

is a solution.

No other real solution exists.

□

5 Complex Extension

The constraint

$$x^2 + y^2 = 0$$

factorizes over the complex numbers:

$$x^2 + y^2 = (x + iy)(x - iy).$$

Hence

$$y = ix$$

or

$$y = -ix.$$

5.1 Complex Reserve-Currency Coordinates

Suppose

$$c_i = a_i + ib_i.$$

The real component

$$a_i$$

represents observable monetary quantities.

The imaginary component

$$b_i$$

represents latent strategic variables.

Examples include:

- Geopolitical influence,
- Military security,
- Strategic reserve value,
- Institutional credibility,
- Future survival probability.

The reserve-currency state becomes

$$\mathcal{C} = (c_1, c_2, c_3, c_4, c_5) \in \mathbb{C}^5.$$

6 Mathematical Admissibility versus Economic Admissibility

A key distinction emerges.

Definition 2. *A state is mathematically admissible if it satisfies the governing equations.*

Definition 3. *A state is economically admissible if it satisfies both the governing equations and the structural constraints of the economic system.*

The complex reserve-currency manifold contains many mathematically admissible states.

Only a subset are economically admissible.

7 The Octic Recoverability Frontier

Suppose recoverable money satisfies an octic constraint

$$M(t) = a_0 + a_1t + a_2t^2 + \cdots + a_8t^8.$$

The recoverability condition may be written abstractly as

$$\Phi_8(M) = 0.$$

Definition 4. *The octic frontier is the boundary separating recoverable monetary trajectories from non-recoverable trajectories under the octic recoverable-money architecture.*

7.1 Interpretation

The octic frontier acts as a conservation law.

Reserve-currency states cannot move arbitrarily throughout the complex manifold.

Instead they must satisfy

$$\Phi_8(M) = 0.$$

The admissible state space becomes

$$\mathcal{M}_{\text{recoverable}} = c \in \mathbb{C}^5 : \Phi_8(M) = 0.$$

8 Loss Aversion and Behavioral Constraints

The complex reserve-currency manifold is symmetric.

Human preferences are not.

Let

$$\lambda > 1.$$

Define

$$U(x) = \begin{cases} x, & x \geq 0, \\ -\lambda|x|, & x < 0. \end{cases}$$

Proposition 1. *Loss aversion destroys rotational symmetry within the reserve-currency manifold.*

Proof. Positive and negative deviations receive different utility weights.

Therefore movement in opposite directions produces unequal welfare effects.

Hence the manifold becomes behaviorally asymmetric.

□

9 Toward a Decic Extension

Suppose recoverable money is generalized to

$$M(t) = a_0 + a_1t + \cdots + a_{10}t^{10}.$$

The recoverability condition becomes

$$\Phi_{10}(M) = 0.$$

This introduces additional degrees of freedom.

9.1 Reserve-Currency Interpretation

A possible interpretation is

$$z = (c_1, c_2, c_3, c_4, c_5)$$

for the octic architecture and

$$z' = (c_1, c_2, c_3, c_4, c_5, c_6)$$

for the decic architecture.

The sixth coordinate may represent:

- Another reserve currency,
- Another nation,
- Another civilization,
- Another autonomous economic bloc,
- Another monetary operating system.

10 Complexity Growth

Pairwise reserve-currency interactions satisfy

$$L_n = \binom{n}{2}.$$

For

$$n = 5,$$

we obtain

$$L_5 = 10.$$

For

$$n = 6,$$

we obtain

$$L_6 = 15.$$

Currencies	Pairwise Links	Increase
5	10	—
6	15	50%

Table 1: Growth of Reserve-Currency Complexity

11 The Recoverability Principle

The central principle emerging from this analysis is:

Do not enlarge the reserve-currency manifold unless recoverability can be demonstrated on the enlarged manifold.

Theorem 3 (Recoverability Principle). *Let*

$$\mathcal{M}_8$$

denote the octic recoverable manifold and

$$\mathcal{M}_{10}$$

denote the decic manifold.

If recoverability is established on

$$\mathcal{M}_8$$

but not on

$$\mathcal{M}_{10},$$

then expansion from

$$\mathcal{M}_8$$

to

$$\mathcal{M}_{10}$$

increases model uncertainty.

Proof. The octic manifold possesses a known recoverability structure.

The decic manifold possesses additional degrees of freedom without an established recoverability guarantee.

Therefore uncertainty increases. □

12 Review

The reserve-currency architecture naturally gives rise to quintic demand and supply structures and a unique real equilibrium under perfect-demand and perfect-supply conditions.

The complex extension reveals a substantially richer reserve-currency geometry. However, recoverability constraints, behavioral asymmetries, institutional limitations, and geopolitical realities dramatically reduce the economically admissible state space.

The octic recoverable-money polynomial therefore serves not merely as a mathematical object but as a solvability frontier separating recoverable monetary architectures from potentially unrecoverable ones.

13 Latent-Factor Monetary Economics

The central hypothesis of this paper is that observable monetary variables are not primitive.

Instead, they are generated by a lower-dimensional latent reserve-currency state.

Define

$$z_t = [z_{1,t} \ z_{2,t} \ z_{3,t} \ z_{4,t} \ z_{5,t}] .$$

The latent coordinates may be interpreted as:

1. Institutional credibility,
2. Fiscal capacity,
3. Military security,
4. Settlement-network centrality,
5. Legal reliability.

Observable reserve-currency premia satisfy

$$p_t = G(z_t).$$

Examples include

$$p_t = [p_{\text{infl}} \ p_{\text{liq}} \ p_{\text{safe}} \ p_{\text{conv}} \ p_{\text{geo}} \ p_{\text{reserve}}].$$

where

- p_{infl} denotes the inflation premium,
- p_{liq} denotes the liquidity premium,
- p_{safe} denotes the safety premium,
- p_{conv} denotes the convenience yield,
- p_{geo} denotes the geopolitical premium,
- p_{reserve} denotes the reserve-currency premium.

Reserve-currency demand becomes

$$M_d = H(z_t).$$

Reserve-currency supply becomes

$$M_s = K(z_t).$$

The reserve-currency system is therefore generated by hidden state variables rather than directly by observable economic quantities.

14 Complex Latent Reserve-Currency States

In the complex extension,

$$z_t = a_t + ib_t.$$

Explicitly,

$$z_t = [a_{1,t} + ib_{1,t} \ a_{2,t} + ib_{2,t} \ a_{3,t} + ib_{3,t} \ a_{4,t} + ib_{4,t} \ a_{5,t} + ib_{5,t}].$$

The real components represent observable monetary quantities.

The imaginary components represent latent strategic quantities.

Possible interpretations include:

- Strategic reserve capacity,
- Geopolitical influence,
- Military capability,
- Institutional resilience,
- Long-term survival probability.

14.1 Observed Projection

The observed monetary system becomes

$$\Re(z_t).$$

The hidden strategic system becomes

$$\Im(z_t).$$

Thus observed reserve-currency statistics reveal only part of the full state.

15 State-Space Dynamics

Suppose policy actions are denoted by

$$u_t.$$

The latent reserve-currency state evolves according to

$$z_{t+1} = F(z_t, u_t, \varepsilon_t).$$

Observed premia satisfy

$$p_t = G(z_t) + \eta_t.$$

where

- ε_t denotes structural shocks,
- η_t denotes observation noise.

15.1 Linear Approximation

Near equilibrium,

$$z_{t+1} = Az_t + Bu_t + \varepsilon_t,$$

and

$$p_t = Cz_t + \eta_t.$$

This yields a standard state-space representation.

16 The Octic Constraint as a State Restriction

The octic recoverability condition

$$\Phi_8(M) = 0$$

acts as a global restriction on state evolution.

Not all trajectories are admissible.

The state must satisfy

$$z_t \in \mathcal{M}_8.$$

where

$$\mathcal{M}_8 = \{z : \Phi_8(M) = 0\}.$$

Definition 5. *The octic manifold is the set of all reserve-currency states consistent with recoverability.*

17 Recoverability as a Conservation Law

In physics, conservation laws restrict motion.

In monetary systems, recoverability restricts admissible state trajectories.

Proposition 2. *The recoverability constraint reduces the effective dimensionality of the reserve-currency state space.*

Proof. Without recoverability,

$$z \in \mathbb{C}^5.$$

This space possesses ten real dimensions.

The constraint

$$\Phi_8(M) = 0$$

removes admissible directions.

Therefore the effective dimensionality is strictly smaller.

□

18 Behavioral Constraints

The complex reserve-currency manifold is mathematically symmetric.

Human preferences are not.

Let

$$\lambda > 1$$

represent loss aversion.

Define

$$U(x) = \begin{cases} x, & x \geq 0, \\ -\lambda|x|, & x < 0. \end{cases}$$

18.1 Behavioral Gradient

The utility gradient is

$$\nabla U.$$

The reserve-currency state therefore experiences a behavioral drift.

Certain directions become preferred.

Others become avoided.

Proposition 3. *Loss aversion further reduces the economically admissible manifold.*

Proof. Movements producing equivalent gains and losses are valued asymmetrically.

Consequently the full state space is no longer behaviorally accessible.

□

19 Governance and Institutional Constraints

Reserve currencies require institutions.

Define

$$K_S = \text{State Capacity.}$$

Then

$$z = f(K_S).$$

State capacity includes:

- Taxation,
- Administration,
- Law enforcement,
- Infrastructure,
- Monetary management.

The reserve-currency state is therefore constrained by governance quality.

19.1 Institutional Feasibility

Not every mathematically admissible reserve-currency state is institutionally feasible.

The feasible set becomes

$$\mathcal{M}_I \subseteq \mathcal{M}_g.$$

20 Military and Strategic Constraints

Reserve currencies possess implicit security backing.

Let

$M = \text{Military Capability.}$

Then

$$p_{\text{safe}} = f(M),$$

and

$$p_{\text{geo}} = g(M).$$

Strategic credibility therefore influences reserve-currency demand.

20.1 Strategic Reserve Value

Let

$q = \text{probability of conflict,}$

and

$L = \text{expected wartime loss.}$

The strategic reserve premium becomes

$$p_{\text{reserve}} = h(qL).$$

Military realities therefore impose additional restrictions on the reserve-currency manifold.

21 Geopolitical Constraints

Reserve currencies operate within international systems.

Define

$H_i = \text{hegemonic influence.}$

Then

$$H_i = f(T_i, M_i, s_i),$$

where

- T_i denotes trade reach,
- M_i denotes military strength,
- s_i denotes reserve share.

The reserve-currency state therefore depends upon geopolitical structure.

22 Reduction of Effective Dimension

We may now identify four successive restrictions.

Initially,

$$z \in \mathbb{C}^5.$$

This yields

10

real dimensions.

Recoverability imposes

$$\Phi_8(M) = 0.$$

Behavior imposes

$$U(x).$$

Institutions impose

$$K_S.$$

Geopolitics imposes

$$H_i.$$

The admissible manifold becomes

$$\mathcal{M}_A = \mathcal{M}_8 \cap \mathcal{M}_B \cap \mathcal{M}_I \cap \mathcal{M}_G.$$

where

- $\mathcal{M}_B$ = behavioral manifold,
- $\mathcal{M}_I$ = institutional manifold,
- $\mathcal{M}_G$ = geopolitical manifold.

Theorem 4. *The economically admissible reserve-currency manifold is generally of substantially lower dimension than the mathematically admissible complex reserve-currency manifold.*

Proof. Each independent constraint removes admissible directions.

The intersection of constraint manifolds therefore possesses lower dimension than the unconstrained state space.

□

23 The Decic Frontier

Suppose the recoverability condition becomes

$$\Phi_{10}(M) = 0.$$

The latent state expands to

$$z' = (c_1, c_2, c_3, c_4, c_5, c_6).$$

The sixth coordinate may represent:

- Another reserve currency,
- Another nation,
- Another civilization,
- Another autonomous monetary bloc.

23.1 Interaction Growth

Pairwise relationships satisfy

$$L_n = \binom{n}{2}.$$

For

$$n = 5,$$

$$L_5 = 10.$$

For

$$n = 6,$$

$$L_6 = 15.$$

Thus

$$L_6 - L_5 = 5.$$

This represents a

$$50\%$$

increase in interaction complexity.

24 Model Uncertainty

The octic system possesses a known recoverability structure.

The decic system may not.

Define

$$U_M = \text{model uncertainty}.$$

Then

$$U_M(\mathcal{M}_{10}) > U_M(\mathcal{M}_8).$$

whenever recoverability remains unproven on the decic manifold.

Proposition 4. *Expansion beyond the octic frontier increases governance risk.*

Proof. Additional dimensions create additional interaction channels.

Unknown recoverability properties increase uncertainty regarding future state evolution.

Therefore governance risk increases.

□

25 Artificial Intelligence Interpretation

The reserve-currency system may be viewed as a constrained representation-learning problem.

Encoder:

$$E : p \rightarrow z.$$

Decoder:

$$D : z \rightarrow \hat{p}.$$

Loss function:

$$L = |p - \hat{p}|^2.$$

Recoverability acts as a regularizer:

$$\Phi_8(M) = 0.$$

The latent state cannot wander arbitrarily.
Instead it must remain on the recoverable manifold.

26 Network Topology of the Octic and Decic Architectures

The reserve-currency system may be represented as a multilayer network.

Define

$$G = (V, E),$$

where

- V denotes vertices,
- E denotes edges.

Vertices represent:

1. Reserve currencies,
2. Nations,
3. Central banks,
4. Settlement institutions,
5. Strategic reserve systems.

Edges represent:

1. Trade relationships,
2. Reserve holdings,
3. Currency swaps,
4. Settlement channels,
5. Strategic dependencies.

26.1 Octic Architecture

Under

$$R_{\max} \leq 5,$$

the reserve-currency layer contains

$$5$$

reserve-currency vertices.

The complete reserve-currency graph possesses

$$L_5 = \binom{5}{2} = 10$$

pairwise relationships.

26.2 Decic Architecture

Under the decic extension,

$$R_{\max} \leq 6.$$

The reserve-currency graph possesses

$$L_6 = \binom{6}{2} = 15$$

pairwise relationships.

The increase is

$$L_6 - L_5 = 5.$$

The proportional increase equals

$$\frac{15 - 10}{10} = 50\%$$

Architecture	Reserve Currencies	Links	Complexity Increase
Octic	5	10	—
Decic	6	15	50%

Table 2: Reserve-Currency Network Complexity

27 Reserve-Currency Interaction Matrices

Define

$$C_5 = (c_{ij}) \in \mathbb{R}^{5 \times 5}$$

for the octic architecture.

Similarly define

$$C_6 = (c_{ij}) \in \mathbb{R}^{6 \times 6}$$

for the decic architecture.

The entries represent:

- Exchange-rate dependencies,
- Reserve substitution effects,
- Settlement relationships,
- Financial contagion channels.

27.1 Spectral Interpretation

The dominant eigenvalue

$$\lambda_{\max}$$

measures systemic influence.

The corresponding eigenvector identifies the reserve-currency structure most responsible for overall network behavior.

28 PGF/TikZ Representation of the Octic Architecture

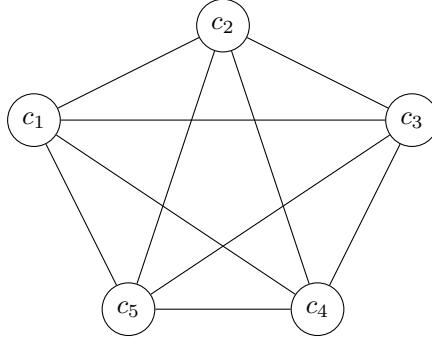


Figure 1: Complete Five-Currency Reserve Network

29 Recoverability as a Control Problem

The reserve-currency system evolves according to

$$z_{t+1} = F(z_t, u_t, \varepsilon_t).$$

The policymaker chooses

$$u_t.$$

The objective is to maintain

$$z_t \in \mathcal{M}_8.$$

29.1 Control Objective

Define

$$J = \sum_{t=0}^T \beta^t U(z_t).$$

The policymaker solves

$$\max_{u_t} J$$

subject to

$$z_t \in \mathcal{M}_8.$$

This transforms recoverability into a constrained optimal-control problem.

30 Welfare Economics of Recoverability

Traditional welfare economics studies:

$$W = W(C).$$

The present framework proposes

$$W = W(C, L, S, R),$$

where

- C = consumption,
- L = liquidity,
- S = safety,
- R = recoverability.

30.1 Recoverability Utility

Let

$$R = \Pr(\text{successful recovery}).$$

Then

$$\frac{\partial W}{\partial R} > 0.$$

Societies value recoverability because future crises are uncertain.

31 Expected Utility under Model Uncertainty

Let

$$\Omega_8$$

denote the octic state space.

Let

$$\Omega_{10}$$

denote the decic state space.

Expected utility becomes

$$EU = \sum_{\omega} p(\omega) U(\omega).$$

Under model uncertainty,

$$p(\omega)$$

is itself uncertain.

31.1 Ambiguity Penalty

Define

$$A = \text{ambiguity penalty.}$$

The effective utility becomes

$$U^* = U - A.$$

The decic architecture may possess larger ambiguity penalties because recoverability is not established.

32 Loss Aversion and the Octic Frontier

Let

$$\lambda > 1$$

measure loss aversion.

Utility satisfies

$$U(x) = \begin{cases} x, & x \geq 0, \\ -\lambda|x|, & x < 0. \end{cases}$$

Proposition 5. *Loss aversion biases policymakers toward the octic architecture.*

Proof. The potential losses associated with unknown decic dynamics are weighted more heavily than equivalent gains.

Hence policymakers prefer architectures possessing established recoverability.

□

33 Political Economy of Expansion

Suppose the sixth reserve-currency coordinate corresponds to

$$c_6.$$

Possible interpretations include:

- Additional nation,
- Additional civilization,
- Additional monetary bloc,
- Additional strategic coalition.

Expansion introduces:

1. New fiscal authorities,
2. New political institutions,
3. New military structures,
4. New settlement systems,
5. New strategic objectives.

Political complexity therefore grows alongside mathematical complexity.

34 International Relations Interpretation

Let

$$H_i = \text{hegemonic influence.}$$

Suppose

$$H_i = f(T_i, M_i, s_i),$$

where

- T_i = trade reach,
- M_i = military capability,
- s_i = reserve share.

The introduction of

$$c_6$$

creates additional strategic relationships.

The resulting international system becomes more difficult to coordinate.

35 Warfare Economics

Reserve currencies possess strategic value.

Let

$$q = \text{Pr}(\text{conflict}),$$

and

$$L = \text{wartime loss.}$$

Strategic reserve value equals

$$qL.$$

The reserve-currency premium satisfies

$$p_{\text{reserve}} = h(qL).$$

Consequently, reserve currencies function not merely as monetary instruments but as strategic assets.

36 Artificial Intelligence and Representation Learning

The reserve-currency architecture may be represented by

Encoder:

$$E : p \rightarrow z.$$

Decoder:

$$D : z \rightarrow \hat{p}.$$

World model:

$$z_{t+1} = W(z_t, u_t).$$

Policy layer:

$$u_t = \pi(z_t).$$

Recoverability regularizer:

$$\Phi_8(M) = 0.$$

Loss function:

$$L = |p - \hat{p}|^2 + \gamma \Phi_8(M)^2.$$

The octic recoverability condition therefore acts as a structural regularizer.

37 Pseudo-Code for Recoverable Monetary Control

Initialize reserve-currency state z

Observe premium vector p

Encode:

```
'''
z = Encoder(p)
'''
```

Check recoverability:

```
'''
Phi8(M)
'''
```

If recoverable:

```
'''
Continue
'''
```

Else:

```

'''
Adjust policy u
'''

Forecast:

'''
z(t+1)=WorldModel(z,u)
'''

Evaluate welfare:

'''
W(C,L,S,R)
'''

Repeat

```

38 A General Recoverability Principle

The analysis suggests the following principle.

Theorem 5 (General Recoverability Principle). *Let*

$$\mathcal{M}_n$$

*denote a monetary manifold generated by a degree- n recoverable-money polynomial.
Expansion from*

$$\mathcal{M}_n$$

to

$$\mathcal{M}_m$$

should occur only if recoverability can be demonstrated on

$$\mathcal{M}_m.$$

Proof. Without recoverability, state evolution cannot be guaranteed to remain within an economically admissible region.

Consequently model uncertainty increases.

Therefore expansion requires recoverability certification.

□

39 The Octic Frontier Hypothesis

The octic architecture occupies a special position.

It simultaneously provides:

1. Finite reserve-currency dimensionality,
2. Recoverability,
3. Computational tractability,
4. Governance feasibility,
5. Strategic stability.

The decic architecture may provide greater flexibility but also introduces additional uncertainty.

40 Final Synthesis

The complete system may be summarized by

$$K_S \rightarrow z \rightarrow p \rightarrow (M_d, M_s) \rightarrow \Phi_8(M) \rightarrow \mathcal{M}_8.$$

where

- K_S = state capacity,
- z = latent reserve-currency state,
- p = observable premium vector,
- M_d = reserve-currency demand,
- M_s = reserve-currency supply,
- $\Phi_8(M)$ = recoverability condition.

The central conclusion is that the economically relevant state space is not the full complex reserve-currency manifold. Instead it is the intersection of:

1. Recoverability constraints,
2. Behavioral constraints,
3. Institutional constraints,
4. Geopolitical constraints,
5. Strategic constraints.

The octic frontier emerges as the boundary beyond which complexity may increase faster than recoverability. Consequently, the decision to introduce an additional reserve-currency dimension, nation, civilization, or monetary bloc is not merely a mathematical question but a question of governance, welfare, strategy, and long-run systemic stability.

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Glossary

$R_{\max}$ Maximum number of reserve-currency dimensions.

z Latent reserve-currency state.

$\mathcal{M}_8$ Octic recoverable manifold.

$\mathcal{M}_{10}$ Decic monetary manifold.

$\Phi_8(M)$ Octic recoverability constraint.

$\Phi_{10}(M)$ Decic recoverability constraint.

K_S State capacity.
 H_i Hegemonic influence.
 p Observed premium vector.
 R Recoverability.
 A Ambiguity penalty.
 L_n Number of pairwise reserve-currency interactions.

The End